The Journal of Open Source Software

MetaCommunityMetrics.jl: A Julia Package for Spatiotemporal Metacommunity Analysis

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 22 October 2025
Published: unpublished

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Summary

Community ecologists are often interested in infering ecological processes from observational data. This involves calculating different metrics such as beta diversity. Calculating these metrics is easy and fast due to the availability of R packages. However, community ecology today is growing beyond analysing single datasets. Large biodiversity databases have expanded the sets of spatio-temporal community data, and there is a growing interest in using simulations to study ecological processes. For these applications, R packages become slow to use.

`MetaCommunityMetrics.jl` is a Julia package that helps ecologists to analyze large datasets of species observations to understand how community composition varies across space and time. By implementing computationally optimized versions of established software and new complementary functions that are designed for handling large spatiotemporal datasets, this package enables ecologists to efficiently analyze large-scale spatiotemporal community datasets that would be computationally prohibitive with existing tools. We designed intuitive interfaces that would feel familiar to ecologists transitioning from R. Our function naming conventions closely match their R equivalents, and we maintained parameter naming and ordering consistency where possible. Some of the metrics we have implemented include beta diversity decompositions in space and time, dispersal-niche continuum index, and variability metric, but a full list of metrics is listed in Table 1. Full documentation and examples, including installation instructions and help files, are available at https://cralibe.github.io/MetaCommunityMetrics.jl/.

Table 1: A summary of the metrics and functions available in `MetaCommunityMetrics.jl`.

Metrics	Functions	Details
Beta diversity decompositions in space and time	`beta_diversity()`	Re-implementation of R package `adespatial`(Dray et al., 2023)
	`spatial_beta_div()`	Re-implementation of R script from Guzman et al. (2022)
	`temporal_beta_div()`	Re-implementation of R script from Guzman et al. (2022)
Dispersal-niche continuum index (DNCI)	`create_groups()`	A novel grouping function for DNCI analysis
	`plot_groups()`	A novel visualization function for DNCI analysis
	`DNCI_multigroup()`	Re-implementation of R script from R package `DNCImper` (Gibert, 2024)
Niche overlap index	`niche_overlap()`	Novel implementation
Occupied patches proportion	`prop_patche()`	Re-implementation of R script from Guzman et al. (2022)

Metrics	Functions	Details
Variability Metric	`CV_meta()`	Re-implementation of R script from Wang et al. (2019)
Niche hypervolume measurements	`MVNH_det()`	Re-implementation from R package `MVNH`(Lu et al., 2021)
	`MVNH_dissimilarity()`	Re-implementation from R package `MVNH`(Lu et al., 2021)
	`average_MVNH_det()`	Novel extension of `MVNH_det()`
	`average_MVNH_dissimilarity()`	Novel extension of `MVNH_dissimilarity()`

Statement of need

Many valuable R (R Core Team, 2024) packages have been developed to aid in community analyses by providing ecological metrics that can summarize processes from biodiversity patterns and have been widely adopted in the field, such as `adespatial` (Dray et al., 2023), `codyn` (Hallett et al., 2020), and `vegan` (Oksanen et al., 2024). However, conducting community analyses in R is often computationally expensive, especially when working with large datasets, a limitation that becomes critical in workflows requiring repeated metric calculations. These include simulation studies for method validation, bootstrap resampling for uncertainty estimation (Efron, 1992), null model approaches for statistical testing (Gotelli & Graves, 1996), and the analysis of large biodiversity databases, such as BioTime, all of which can create significant computational bottlenecks when processing multiple large datasets iteratively. To improve the efficiency of these computationally intensive analyses, `Julia` becomes a logical choice for re-implementations. `Julia` is known for its just-in-time compilation that optimizes code execution to levels comparable with lower-level languages like C or Fortran (Bezanson et al., 2017), yet `Julia` remains relatively uncommon in ecological workflows.

State of the field

To position `MetaCommunityMetrics.jl` within the `Julia` ecosystem, we conducted a comprehensive review of existing `Julia` packages under the EcoJulia organization, a community-driven effort providing tools for ecological and environmental analysis, and the PoisotLab, which develops many `Julia` packages focusing on quantitative and computational ecology. We ensured that our package builds beyond the tools currently existing in `Julia` even when sharing similar ecological concepts. For example, our beta diversity metrics share the concept of beta diversity with `Diversity.jl` (Robert Reeve & Harris, 2024). However, our implementation decomposes beta diversity into species replacement and richness differences (or abundance difference when abundance data is used instead of occurrence data) (Baselga, 2010, 2012, 2013; Legendre, 2014), which differentiates our approach from `Diversity.jl`, which emphasizes alpha, beta, and gamma diversity measures at the metacommunity and community levels (Richard Reeve et al., 2014). To our knowledge, no existing `Julia` packages implement the specific metrics listed in Table 1. `MetaCommunityMetrics.jl` addresses this gap by providing the first `Julia` implementations of these metrics, which are essential for research in community ecology but were previously only available through computationally expensive R implementations.

Software Design

We built `MetaCommunityMetrics.jl` on three design principles: computational efficiency, usability for spatiotemporal data, and methodological validation.

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First, we extended single-snapshot metrics (single site and/or time point) to operate on multidimensional datasets (multiple sites and/or time points) via aggregation functions (`spatial_beta_div()`, `temporal_beta_div()`, `average_MVNH_det()`, `average_MVNH_dissimilarity()`), trading complexity for usability and eliminating manual looping.

Second, we prioritized performance over comprehensive coverage (compared to the `R` implementations). Our beta diversity focuses on the Podani family approach rather than all five coefficient families, `MVNH_dissimilarity` implements only Bhattacharyya distance, and DNCI lacks built-in visualizations. We plan to expand these in future releases.

Third, we use vectorized operations where possible. We also employ Cholesky decomposition for covariance calculations (a computational optimization over LU decomposition in the original R implementation), providing substantial performance gains for positive-definite matrices (Haddad, 2024).

Fourth, our DNCI implementation accepts empty sites and singletons using Zero-adjusted Bray-Curtis dissimilarity, filters uninformative species (completely absent or ubiquitous), and implements status flags for five edge cases rather than throwing errors or returning NAs, supporting simulation workflows where sparse matrices are meaningful. We developed `create_groups()` and `plot_groups()` to automate site grouping (required by Vilmi et al. (2021) but manual in R) and visualize results. See documentation for details.

Finally, all implementations were validated against R equivalents to ensure methodological consistency (see documentation).

Research Impact Statement

`MetaCommunityMetrics.jl` delivers substantial computational advantages over existing tools. For example, we re-implemented the `beta.div.comp` function from the `R` package `adespatial` (Dray et al., 2023), achieving up to $\sim 58\times$ faster execution times while maintaining methodological consistency (see Figure 1). As shown in Figure 1, performance gains range from $\sim 3\times$ to $58\times$ across all re-implemented functions using three testing datasets (large: 53,352 observations; medium: 26,676 observations; small: 5,325 observations), positioning `MetaCommunityMetrics.jl` as an efficient solution for diverse ecological research needs, from standard biodiversity analyses to computationally intensive simulation studies and large-scale data processing.

Further, the first empirical application of Approximate Bayesian Computation methods to connect metacommunity theory with observed data (a long-term phytoplankton community monitoring data) is using this package to calculate summary statistics. The manuscript of this application is under preparation.

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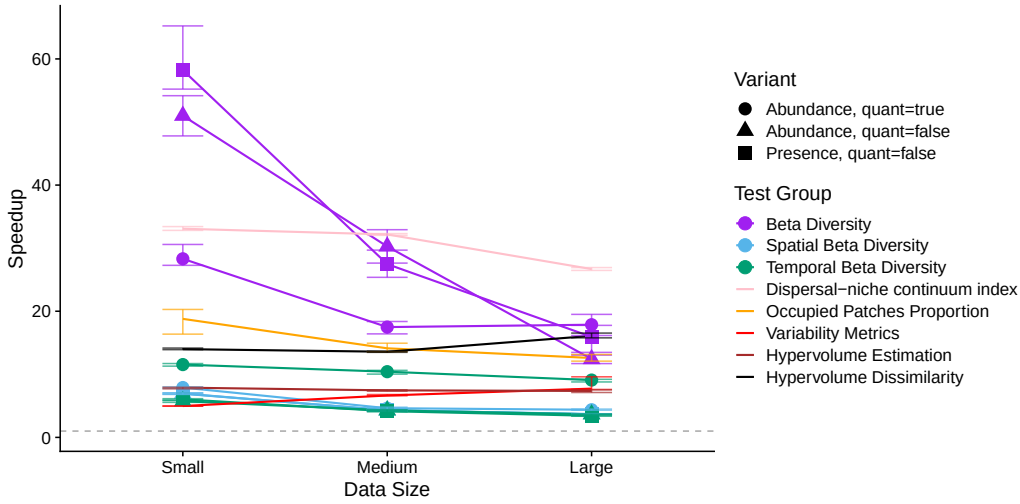

Figure 1: A plot showing the speedup of all benchmarked functions across three datasets: small (5,335 observations), medium (26,676 observations), and large (53,352 observations). Speedup is calculated as the `R` median execution time divided by the `Julia` median execution time. Median speedup and its confidence interval of each comparison is shown. The grey dashed line shows where speedup equals to 1, that is both `R` and `Julia`'s implementations require the same amount of time. The calcualation of beta diversity has two variants: `quant = true` (abundance data is used) and `quant = false`(occurrence data is used). Details about this benchmarking can be found in the documentation under the benchmark results section.

AI usage disclosure

We used Claude, a generative AI tool, for proofreading and identifying grammatical errors in the manuscript and software documentation, and for providing optimization suggestions during software development. All writing suggestions were manually reviewed by the authors and selectively accepted. For optimization suggestions during software development, all suggestions were manually reviewed by the authors and selectively accepted, verified to produce the same results as before optimization, and benchmarked to confirm improvements in execution time and/or memory usage.

Acknowledgement

We are grateful to the authors of the pre-existing `R` packages/implementations that we have implemented in `MetaCommunityMetrics.jl`. We express our gratitude to everyone who has tried out this package and provided feedback on how to improve it. LMG acknowledges funding from USC and Cornell University, YYJC acknowledges funding from the Wrigley Institute.

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